LESSON GUIDE

Technology Plus

Unit 3: Hardware

2.5.2 – Peripheral Devices

Lesson Overview:

In this lesson, students will compare different input and output interfaces for peripheral devices.

Students will:

- Compare input and output interfaces for peripheral devices.
- Analyze scenarios and determine which peripheral interface best meets the user's needs.

Guiding Question: What are the different input and output interfaces for peripheral devices and how do they work?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 50-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 2.5 – Compare and contrast common types of input/output device interfaces.
  - Peripheral devices
    - USB (A/B/C)
    - Thunderbolt
    - Bluetooth
    - Radio frequency (RF)
    - Lightning

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Lesson Background

Background Information:

Peripheral interfaces are the connection points that allow computers to communicate with external devices, these interfaces provide pathways for data transfer and power supply. The most widely used interface is the universal serial bus (USB). It comes in different versions and connector types such as A, B, and C. Other proprietary or specialized wired interfaces like Lightning and Thunderbolt offer high-speed data transfer and advanced features for specific systems or device setups. Wireless interfaces such as Bluetooth and radio frequency are common for peripherals where mobility is needed.

By the end of the lesson, students should be able to evaluate a scenario and determine which peripheral interface is needed.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 2.5.2 – Peripheral Devices • Activity Slides 2.5.2 – Peripheral Interface Recommendation
Materials per Student	• Student Notes 2.5.2 – Peripheral Devices • Student Handout 2.5.2 – Peripheral Devices • Assessment 2.5.2 – Peripheral Devices (optional)

Lesson Background

Cyber Terms:

Term	Definition
Universal Serial Bus (USB)	a standardized interface used to connect devices to a computer or other host systems for data transfer and/or power supply
Thunderbolt	a high-speed interface that uses the USB-C connector, it supports data transfer, video output, and power delivery
Bluetooth	a short-range wireless technology used for connecting devices over relatively short distances
radio frequency (RF)	a wireless technology that uses radio waves to communicate; used in various peripherals like wireless keyboards, mice, and gaming controllers
Lightning	a proprietary connector developed by Apple, used in various Apple devices like iPhones, iPads, and some of their accessories

2.5.2 – Peripheral Devices

Guiding Question: What are the different input and output interfaces for peripheral devices and how do they work?

Lesson Launch (5-10 minutes)

- Peripheral interfaces provide the connection between a device and its peripheral devices so they can exchange data and/or power.
- Peripheral devices are external devices that connect to a computer to add features, improve performance, or provide additional input and output options.
- Peripheral interfaces:
    - Enable communication between a device and connected peripherals to send and receive data.
    - Supply power to charge or run the device it is connected to.
    - Define a common way for devices to connect with minimal compatibility issues.
    - Can support high-speed data and video or provide wireless convenience.
- Students learn more about peripheral devices in lesson 2.4.1, which should be taught before this lesson.

Peripheral Interfaces (20-25 minutes)

- Distribute Student Handout 2.5.2 – Peripheral Devices and encourage students to record their answers during the lesson.
- Discuss universal serial bus (USB), its definitions, and different types.
    - USB replaces older interfaces like serial and parallel ports; it simplifies the process of connecting peripherals to devices.
    - USB-A is universally supported by most computers and devices.
    - Older versions of USB-A have lower data transfer rates compared to newer versions.
    - USB-A is larger and less versatile in comparison to newer USB types.
    - USB-B is often used in devices where the connector is not frequently plugged and unplugged.
    - USB-B is rare in modern devices.
    - USB-B has a larger connector in comparison to newer USB types.
    - USB-C supports high data transfer rates.
    - USB-C can deliver higher power for charging devices.
    - Not all devices have adapted USB-C yet.
- Introduce Thunderbolt, its definition, and discuss its features:

  - Thunderbolt supports data transfer rates up to 40 Gbps.
  - Thunderbolt can connect multiple types of devices, including displays and external drives, through a single port.
  - Thunderbolt is typically more expensive due to advanced technology.
- Introduce Bluetooth, its definition, and discuss its features:
  - Bluetooth connects devices such as keyboards, mice, and headphones to other devices.
  - Bluetooth is designed to use low power and is highly compatible with battery-operated devices.
  - Bluetooth has generally slower data transfer rates compared to wired connections.
- Introduce radio frequency, its definition, and discuss its features:
  - RF includes technologies like Wi-Fi, Bluetooth, and near-field communication.
  - RF operates on different frequencies based on the technology it is being used with, allowing it to reduce interference.
  - Devices with long-range RF systems have a high-power consumption, while those with short-range communication have a low-power consumption.
  - Bluetooth is an example of RF technology, but RF is not an example of Bluetooth.
- Introduce Lightning, its definition, and discuss its features:
  - Lightning is optimized for use with Apple devices only and supports both data transfer and charging.
  - Lightning is a small and reversible connector.
  - Lightning accessories and cables tend to be more expensive

Activity: Peripheral Recommendation (20-25 minutes)

- Display Activity Slides 2.5.2 – Peripheral Recommendation and have students record answers on the handout.
- Decide if students will work individually or in small groups.
- Read each scenario aloud (or allow students/groups to read independently).
- Students determine which peripheral interface best meets the user's needs.
- Allow time for students to write their answers.
- Remind students that answers may vary, as long as they are justified and reasonable..

Scenarios and Answers

- Jordan is a freelancer who edits 4k videos on his laptop. He needs to connect an external monitor, an SSD, and keep his laptop charged for long work sessions. Which interface would handle these tasks effectively?

  Thunderbolt is the best choice of interface because it can support data transfer, high-resolution video output, and power delivery through one cable.

- A company issues iPhones to employees and needs a standard cable for transferring media. Which interface should the company use?

  Lightning connectors are the only interface that are compatible with iPhones only.

- Alex is a competitive gamer who wants a wireless keyboard and mouse but needs the lowest latency possible. Which interfaces should she choose?

  Alex should choose a radio frequency interface because they provide near-wired performance with very low latency. Although Bluetooth is wireless, it has a higher latency and could experience interference from other devices.

- A company wants to give all employees wireless headphones for virtual meetings. Which interface should they choose for easy connectivity and compatibility?

  Bluetooth is the best choice for easy connectivity and compatibility because it is built into most laptops, phones, and tablets. It allows quick pairing and works with multiple devices.

- Chris needs to connect an office printer to his desktop computer. The printer does not support wireless connections and only has a standard port. Which interface should Chris use to make the connection?

  Chris should use a USB-A and USB-B interface because they are compatible with most computers and devices. USB-B plugs into the port on the back of the printer, and the USB-A end connects to an available port on the computer.

Putting It All Together (5-10 minutes)

- Review what students learned today by discussing the questions below as a class.
- What is the purpose of peripheral interfaces?

  Peripheral interfaces provide the connection between a device and its peripherals so they can exchange data and/or power. Peripheral interfaces define a common way for devices to connect with minimal compatibility issues, can support high-speed data and video, and provide wireless convenience.

- What should a user consider when choosing a USB Type?

  When choosing a USB, a user should consider what devices they are connecting to and how fast they need data to transfer.

- How can someone distinguish between Bluetooth and radio frequency?

  Radio frequency (RF) communication is a broad category that includes technologies such as Wi-Fi, near-field communication, and Bluetooth. Each operates on different frequencies to help reduce interference. Bluetooth, however, is a short-range RF technology that is more prone to interference and offers a shorter range compared to others